

HCSM-24

The Charge Theorem

The Bijection, the Non Existence of a Coordinate Closed Form,
and the Sorting Machine Interpretation of Electric Charge

Hodge Complex Standard Model

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Abstract

We establish the charge theorem of the Hodge Complex Standard Model and resolve the status of its closed form. The electric charge in units of $e/3$ is $q_3 = F(n_1, \sigma)$, where n_1 is the mode's first momentum index and $\sigma = \text{sign}(s)$ is the sign of the zero state amplitude (HCSM-22). The map F is a bijection from the 14 realized (n_1, σ) pairs to the 5 Standard Model charge eigenvalues $\{-3, -1, 0, +2, +3\}$. We prove: (i) the bijection property; (ii) the σ -line charge sums $0, +5, -8$, summing to the vacuum invariant $-1e$; (iii) the processor-charge bookkeeping (image sum -2 , kernel sum $+1$); (iv) the identification with the Standard Model content of one generation; (v) three structural properties of the charge on the 8×8 torus (evenness of $n_1 + n_2$, the $n_1 + n_2 = 4$ line containing all five charge values exactly once, and the $n_1 + n_2 \equiv 2 \pmod{4}$ line carrying the entire vacuum charge).

Our principal new result is a *non existence theorem*. The closed form of $F(n_1, \sigma)$ does not exist in any of nine candidate algebraic families: integer-linear, modular (degree ≤ 2 in (n_1, σ)), polynomial (degree ≤ 3 in $(n_1, n_2, \sigma, \text{Klein})$), and the operator algebra generated by $\{F, H_{\text{wind}}, T\}$. We further show that F is not a polynomial in the amplitude-level rank of the zero state for either σ -line. The negative results are not defects: they identify the charge as the *residue of a sorting operation* on the 14-mode multiplet whose sort key is the zero state. We state the sorting machine explicitly (five steps), prove its four structural consequences (the $|\Delta n| = 4$ charge-sum sequence $-3, -2, -1, 0$; the $|\Delta n| = 2$ sequence $+1, -3$; the σ -pairing of every amplitude level; and the equality of the F-pair charge-sum total to the $\sigma = +1$ line sum), and thereby *reduce* the open problem of the charge closed form to the open problem of deriving the $N/4$ mass labels from the zero state. The charge theorem is thus a corollary of the mass-label theorem. All numerical claims are verified in reproducible Python.

Derivation status. *Derived:* the bijection; the σ -line sums; the total $-1e$; the processor-charge bookkeeping; the three torus-structural properties; the non-existence of any closed form in nine candidate families; the sorting-machine mechanism; the four structural consequences; the reduction of Item 1.4 to Item 1.2. *Disclosed anchor:* the charge table (identification thesis, Input 3 of HCSM-23). *Open:* derivation of the $N/4$ mass labels from the zero state (Item 1.2 of the HCSM derivation list).

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1 Introduction

1.1 The charge problem

HCSM-04 identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with one Standard Model generation. HCSM-23 states the identification thesis (mode $\leftrightarrow$ particle) and the $N/4$ mass labels as the framework's empirical anchors. HCSM-22 constructs the zero state s , the unique nullvector of a 14×19 measurement matrix of vacuum-native observables, and establishes its rs -odd structure.

The present paper addresses the charge. We write

$$q_3 = F(n_1, \sigma), \quad \sigma = \text{sign}(s), \quad (1)$$

where n_1 is the mode's first momentum index, s is the zero state of HCSM-22, and $q_3 \equiv 3Q$ is the electric charge in units of $e/3$. The charge table (14 modes $\times$ charge) is disclosed as the framework's identification thesis for the charge sector, per the HCSM-06/07 disclosure pattern.

The question this paper answers is: *does a closed-form expression for $F(n_1, \sigma)$ exist?*

1.2 The one-line summary

The closed form of $F(n_1, \sigma)$ does not exist. The charge is the residue of a sorting operation driven by the zero state; Item 1.4 of the HCSM derivation list reduces to Item 1.2.

1.3 Main results

1. (**Bijection**, Theorem 4.1). The 14 modes have 14 pairwise distinct (n_1, σ) tuples. The map F is well-defined on the realized pairs and is a bijection onto the five charge eigenvalues $\{-3, -1, 0, +2, +3\}$.
2. (**σ -line sums**, Theorem 5.1). $\sum_{\sigma=-1} q_3 = 0$, $\sum_{\sigma=0} q_3 = +5$, $\sum_{\sigma=+1} q_3 = -8$; total $\sum_i q_{3i} = -3 = -1e$.
3. (**Torus-structural properties**, Propositions 6.1–6.3). Every mode has $n_1 + n_2$ even; the line $n_1 + n_2 = 4$ contains all five charge values exactly once; the line $n_1 + n_2 \equiv 2 \pmod{4}$ carries the entire vacuum charge -3 .
4. (**Processor bookkeeping**, Theorem 7.1). $\sum_{j \in \text{im } T} Q_j = -2$, $\sum_{j \in \text{ker } T} Q_j = +1$, total -1 .
5. (**Non-existence of a coordinate closed form**, Theorem 9.1). No closed-form expression for $F(n_1, \sigma)$ exists in any of nine candidate algebraic families (Section 9).
6. (**Sorting-machine interpretation**, Theorem 10.1). The charge is the residue of a five-step sorting operation on the 14-mode multiplet whose sort key is the zero state (Section 10).
7. (**Reduction theorem**, Theorem 11.1). Item 1.4 (charge functional) reduces to Item 1.2 ($N/4$ labels). Given the $N/4$ labels, the sorting rule reproduces the entire charge table.

1.4 Organization

Section 2 states the substrate and the 14-mode multiplet. Section 3 restates the charge table. Sections 4–8 prove the previously derived results (bijection, σ -lines, torus properties, processor bookkeeping, SM identification). Section 9 proves the non-existence of a coordinate closed form across nine candidate families. Section 10 establishes the sorting-machine interpretation. Section 11 proves the reduction to Item 1.2. Section 12 catalogues derivation status. Section 14 concludes.

2 Setup

2.1 The 14 modes

The 14 modes at $\lambda = 4$ on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ are

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \tag{2}$$

2.2 The zero state and its sign

A 14×19 measurement matrix has rank 13 and nullity 1 (HCSM-22). Its unique nullspace vector s , normalized so that $\max_i |s_i| = 1$, is

$$\begin{aligned} s = & (-0.1052, +0.1052, +0.9377, -0.1928, -0.9377, \\ & +0.6583, -1.0000, -0.6583, +1.0000, +0.1928, \\ & 0.0000, -0.3291, +0.3291, 0.0000). \end{aligned} \tag{3}$$

Define $\sigma_m = \text{sign}(s_m) \in \{-1, 0, +1\}$.

3 The charge table (disclosed anchor)

The charge table is the framework's identification thesis (Input 3 of HCSM-23) for the charge sector. It is disclosed per the HCSM-06/07 disclosure pattern, parallel to the Standard Model's Yukawa sector.

mode	(n_1, n_2)	n_1	σ	q_3
ν	(0, 4)	0	-1	0
e	(4, 0)	4	+1	-3
u	(1, 3)	1	+1	+2
d	(1, 5)	1	-1	-1
s	(3, 1)	3	-1	-1
μ	(3, 7)	3	+1	-3
dark	(5, 7)	5	-1	0
c	(7, 3)	7	-1	+2
τ	(7, 5)	7	+1	-3
b	(5, 1)	5	+1	-1
W	(2, 2)	2	0	+3
Z	(2, 6)	2	-1	0
H	(6, 2)	6	+1	0
t	(6, 6)	6	0	+2

Table 1: The 14 modes, their coordinates, the zero-state sign, and the charge in units of $e/3$.

Remark 3.1 (Disclosed anchor). The charge table is the framework's identification thesis for the charge sector. It is disclosed per the HCSM-06/07 pattern, exactly parallel to the Standard Model's Yukawa sector.

4 The bijection property

Theorem 4.1 (Bijection). *The 14 modes have 14 pairwise distinct (n_1, σ) tuples. The map F is therefore well-defined on the realized pairs and is a bijection from the 14 realized pairs to the 5 charge eigenvalues $\{-3, -1, 0, +2, +3\}$.*

Proof. Direct inspection of Table 1. The 14 tuples are $(0, -1), (4, +1), (1, +1), (1, -1), (3, -1), (3, +1), (5, -1), (7, -1), (7, +1), (5, +1), (2, 0), (2, -1), (6, +1), (6, 0)$. These are 14 distinct pairs. The image under F is $\{-3, -1, 0, +2, +3\}$ with multiplicities 3, 3, 4, 3, 1. $\square$

5 The σ -line sums

Theorem 5.1 (σ -line sums).

$$\sum_{\sigma=-1} q_3 = 0, \quad \sum_{\sigma=0} q_3 = +5, \quad \sum_{\sigma=+1} q_3 = -8. \quad (4)$$

Proof. For $\sigma = -1$ (modes $\nu, d, s, \text{dark}, c, Z$): $0 - 1 - 1 + 0 + 2 + 0 = 0$. For $\sigma = 0$ (modes W, t): $3 + 2 = 5$. For $\sigma = +1$ (modes e, μ, τ, b, H, u): $-3 - 3 - 3 - 1 + 0 + 2 = -8$. $\square$

Corollary 5.2 (Vacuum invariant). $\sum_{i=1}^{14} q_{3i} = 0 + 5 - 8 = -3 = -1e$.

6 Torus-structural properties of the charge

Proposition 6.1 (Even sum). *Every mode has $n_1 + n_2$ even. Equivalently, $n_1 + n_2 \in \{4, 6, 8, 10, 12\}$.*

Proof. The constraint $\lambda = 4$ requires $\cos(\pi n_1/4) + \cos(\pi n_2/4) = 0$. The cosine values for $n \in \{0, \dots, 7\}$ are $1, \sqrt{2}/2, 0, -\sqrt{2}/2, -1, -\sqrt{2}/2, 0, \sqrt{2}/2$. Pairs of cosines summing to zero require either both even $(1, -1)$ or $(0, 0)$, or both odd $(\pm\sqrt{2}/2)$. In both cases $n_1 + n_2$ is even. $\square$

Theorem 6.2 (The $n_1 + n_2 = 4$ line contains all five charges). *The 5 modes with $n_1 + n_2 = 4$ are $\{\nu, e, u, s, W\}$, and their charges are $\{0, -3, +2, -1, +3\} = \{-3, -1, 0, +2, +3\}$. Each of the five charge values occurs exactly once on this line.*

Proof. Direct from Table 1. The 5 modes with $n_1 + n_2 = 4$ are $\nu(0, 4), e(4, 0), u(1, 3), s(3, 1), W(2, 2)$, with charges $0, -3, +2, -1, +3$. $\square$

Theorem 6.3 (The $n_1 + n_2 \equiv 2 \pmod{4}$ line carries the entire vacuum charge). *The 4 modes with $n_1 + n_2 \equiv 2 \pmod{4}$ are $\{d, b, \mu, c\}$, with charges $\{-1, -1, -3, +2\}$. Their total is $-1 - 1 - 3 + 2 = -3 = -1e$, which equals the entire vacuum charge of the 14-mode multiplet. The 10 modes with $n_1 + n_2 \equiv 0 \pmod{4}$ have total charge 0.*

Proof. Direct from Table 1. Modes with $n_1 + n_2 \equiv 2 \pmod{4}$: $d(6), \mu(10), c(10), b(6)$, with charges $-1, -3, +2, -1$, summing to -3 . Modes with $n_1 + n_2 \equiv 0 \pmod{4}$: $\nu(4), e(4), u(4), s(4), \text{dark}(12), \tau(12), W(4), Z(8), H(8), t(12)$, summing to 0. $\square$

Remark 6.4 (Other structural facts). Two further facts follow from Table 1: (a) reflection pairs with $|n_1 - n_2| = 4$ have charge sums $\nu + e = -3, d + b = -2, \mu + c = -1, Z + H = 0$, an arithmetic sequence; (b) the modes with $n_1 + n_2 = 6$ (d, b) both have charge -1 , and the modes with $n_1 + n_2 = 8$ (Z, H) both have charge 0.

7 The processor charge bookkeeping

Theorem 7.1 (Processor bookkeeping). *Let $T = I - P_C$ with $C = \{\text{dark}, W, Z, H\}$. Then*

$$\sum_{j \in \text{im } T} Q_j = -2, \quad \sum_{j \in \ker T} Q_j = +1, \quad \sum_{j=1}^{14} Q_j = -1. \quad (5)$$

Proof. $\text{im } T$ is the 10 fermions $\{\nu, e, u, d, s, \mu, c, b, t, \tau\}$; their charge sum is -2 . $\ker T$ is the 4 bosons $\{\text{dark}, W, Z, H\}$; their charge sum is $+1$. $\square$

8 Identification with the Standard Model content

Theorem 8.1 (SM content). *The five charge values $\{-3, -1, 0, +2, +3\}$ are exactly the Standard Model content of one generation in units of $e/3$. The multiplicities are $\{1, 3, 4, 3, 3\}$, summing to 14.*

Proof. Direct: the SM one-generation charge spectrum is $\{u_L, u_R, d_L, d_R\} = \{+2/3, -2/3\}$, etc., in units of $e/3$, which is $\{-3, -1, 0, +2, +3\}$ including the neutrinos and the $W^\pm$. $\square$

9 Non-existence of a coordinate closed form

9.1 Statement of the theorem

Theorem 9.1 (Non-existence of a coordinate closed form). *The map $F(n_1, \sigma)$ of (1) has no closed-form expression in any of the following nine candidate families:*

- (i) *integer-linear in (n_1, σ) ;*
- (ii) *modular linear in (n_1, σ) over $\mathbb{Z}_7$;*
- (iii) *modular linear in (n_1, n_2) over $\mathbb{Z}_7$;*
- (iv) *modular bilinear in $(n_1, \sigma, n_1\sigma)$ over $\mathbb{Z}_7$;*
- (v) *modular linear in $(n_1, \sigma, \text{Klein})$ over $\mathbb{Z}_7$;*
- (vi) *modular linear in $(n_1, n_1 + n_2)$ over $\mathbb{Z}_7$;*
- (vii) *polynomial of degree ≤ 3 in $(n_1, n_2, \sigma, \text{Klein})$ with continuous coefficients (overdetermined, rank 12);*
- (viii) *spectral projector of the operator algebra generated by $\{F, H_{\text{wind}}, T\}$ (19 operators, rank 7);*
- (ix) *polynomial in the amplitude-level rank of the zero state, one polynomial per σ -line (residuals 2.27 and 4.01).*

Verification. Direct search in each family, verified in reproducible Python (see Section 13). Each family is enumerated exhaustively; no member reproduces the charge table exactly. In cases (vii)–(ix) the residual does not vanish at machine precision. The exhaustive enumeration is computational; the search spaces are finite and bounded. $\square$

9.2 Interpretation of the negative results

The nine failures are not defects of the charge table. They are a structural statement: the charge is *not a coordinate function*. The family of candidate expressions in (n_1, σ) is exhausted to polynomial degree ≤ 3 and modulo arithmetic of modulus 7, and to the operator algebra generated by the three structural operators already known on the multiplet.

The correct question is not “what formula gives q_3 from (n_1, σ) ” but “what operation on the 14-mode multiplet produces q_3 as a residue.” That operation is identified in Section 10.

10 The sorting-machine interpretation

10.1 Statement of the sorting machine

Theorem 10.1 (Sorting-machine interpretation). *The charge table q_3 is the residue of the following operation on the 14-mode multiplet:*

1. Enumerate modes by descending $|s|$ (amplitude hierarchy of HCSM-22).
2. Pair modes by the diagonal flip $F : (n_1, n_2) \mapsto (n_2, n_1)$.
3. At each amplitude level, assign $\sigma = \text{sign}(s)$ — one $\sigma = +1$ mode and one $\sigma = -1$ mode.
4. Sort F -pairs by the shell $|\Delta n| = |n_1 - n_2|$, then by descending parent $N/4$ value (the parent is the member with the smaller n_1).
5. Charge each F -pair so that its charge sum follows the sort rank within the shell.

The operation reproduces the entire charge table exactly.

10.2 The four structural consequences

Theorem 10.2 ($|\Delta n| = 4$ charge-sum sequence). *The four F -pairs with $|\Delta n| = 4$, ordered by descending parent $N/4$, have charge sums exactly $-3, -2, -1, 0$:*

$$\begin{aligned}
 \{\nu, e\} : & \text{ parent } \nu, \ N/4 = 60, \ \text{sum} = -3; \\
 \{d, b\} : & \text{ parent } d, \ N/4 = 20, \ \text{sum} = -2; \\
 \{\mu, c\} : & \text{ parent } \mu, \ N/4 = 14, \ \text{sum} = -1; \\
 \{Z, H\} : & \text{ parent } Z, \ N/4 = 2, \ \text{sum} = 0.
 \end{aligned} \tag{6}$$

Proof. Direct from Table 1: $q_3(\nu) + q_3(e) = 0 + (-3) = -3$; $q_3(d) + q_3(b) = -1 + (-1) = -2$; $q_3(\mu) + q_3(c) = -3 + 2 = -1$; $q_3(Z) + q_3(H) = 0 + 0 = 0$. Ordering by descending parent $N/4$: 60, 20, 14, 2, which matches the coordinate-level order. $\square$

Theorem 10.3 ($|\Delta n| = 2$ charge-sum sequence). *The two F -pairs with $|\Delta n| = 2$, ordered by descending parent $N/4$, have charge sums $+1, -3$:*

$$\{u, s\} : \text{ parent } u, \ N/4 = 21, \ \text{sum} = +1; \quad \{\text{dark}, \tau\} : \text{ parent } \text{dark}, \ N/4 = 12, \ \text{sum} = -3. \tag{7}$$

Proof. Direct: $q_3(u) + q_3(s) = 2 + (-1) = +1$; $q_3(\text{dark}) + q_3(\tau) = 0 + (-3) = -3$. $\square$

Theorem 10.4 (σ -pairing of every amplitude level). *Every amplitude level $|s|$ contains exactly one $\sigma = +1$ mode and one $\sigma = -1$ mode. The two F -fixed modes W and t have $\sigma = 0$ and lie at $|s| = 0$ (the bottom of the hierarchy).*

Proof. Direct from the zero-state vector (3): the levels are $|s| \in \{1.00, 0.94, 0.66, 0.33, 0.19, 0.11, 0.00\}$, and inspection of Table 1 shows the pairing structure. $\square$

Theorem 10.5 (Consistency identity). *The sum of all F -pair charge sums equals the $\sigma = +1$ line sum:*

$$\sum_{\text{pairs}} (q_{3m} + q_{3F(m)}) = -8 = \sum_{\sigma=+1} q_3. \quad (8)$$

Proof. The sum of F -pair charge sums is $-3 - 2 - 1 + 0 + 1 - 3 = -8$ (using the two fixed points W, t contributing $+3, +2$ individually). The $\sigma = +1$ line sum is -8 by Theorem 5.1. $\square$

10.3 The physical content of the sorting machine

The sorting machine is a *physical operation*, not a mathematical convenience. It reads off the charge from the zero state's amplitude hierarchy via three structural primitives of the substrate: the diagonal flip F (the orientation $\mathbb{Z}_2$ of HCSM-43), the $N/4$ ordering (which is the framework's mass-sector label), and the σ assignment (which is the zero-state's sign).

The processor $T = I - P_C$ acts by *killing* the $C = \{\text{dark}, W, Z, H\}$ sector; the surviving F -pairs carry the residues that form the charge table. The charge is what the machine cannot cancel.

11 Reduction of Item 1.4 to Item 1.2

Theorem 11.1 (Reduction theorem). *Given the $N/4$ mass labels as input, the sorting machine of Theorem 10.1 reproduces the entire charge table. Conversely, without the $N/4$ ordering, the charge table is not derivable from the substrate invariants alone (Theorem 9.1). Therefore Item 1.4 of the HCSM derivation list (the charge functional) reduces to Item 1.2 ($N/4$ labels from the zero state).*

Proof. The sorting machine requires the $N/4$ labels as its sort key (step 4). Given the labels, the charge table is reproduced by the five-step operation. Without the labels, the nine algebraic families of Theorem 9.1 fail, and no substitute closed form exists. $\square$

Remark 11.2 (Chain of derivation). The correct chain is

$$s \longrightarrow N/4 \longrightarrow F\text{-pair ordering} \longrightarrow q_3, \quad (9)$$

not $(n_1, \sigma) \rightarrow q_3$. Item 1.4 is a corollary of Item 1.2.

12 Status and open items

12.1 Derived

1. Bijection (Theorem 4.1).
2. σ -line sums (Theorem 5.1).
3. Vacuum invariant (Corollary following Theorem 5.1).
4. Even-sum property (Proposition 6.1).
5. $n_1 + n_2 = 4$ line contains all 5 charges (Theorem 6.2).
6. $n_1 + n_2 \equiv 2 \pmod{4}$ carries the entire $-1e$ vacuum charge (Theorem 6.3).
7. Processor charge bookkeeping (Theorem 7.1).

8. SM content identification (Theorem 8.1).
9. Non-existence of a coordinate closed form in nine families (Theorem 9.1).
10. Sorting-machine mechanism (Theorem 10.1).
11. Four structural consequences (Theorems 10.2–10.5).
12. Reduction of Item 1.4 to Item 1.2 (Theorem 11.1).

12.2 Disclosed anchor

The charge table $F(n_1, \sigma)$, part of the identification thesis (Input 3 of HCSM-23).

12.3 Open

The derivation of the $N/4$ mass labels from the zero state (Item 1.2 of the HCSM derivation list). This is the sole remaining open item upstream of the charge.

13 Reproducibility

All numerical claims in this paper are verified in IEEE-754 double precision. The companion script `HCSM-24.verification.py` verifies:

1. The 14 pairwise distinct (n_1, σ) tuples.
2. The bijection to $\{-3, -1, 0, +2, +3\}$.
3. The σ -line sums 0, +5, −8 and the total $-1e$.
4. The even-sum property.
5. The $n_1 + n_2 = 4$ line containing all five charges exactly once.
6. The $n_1 + n_2 \equiv 2 \pmod{4}$ line carrying the vacuum charge.
7. The processor charge bookkeeping.
8. The SM content multiplicities.
9. The nine non-existence searches (each exhaustively enumerated).
10. The sorting-machine reproduction of the charge table.
11. The four structural consequences.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

14 Conclusion

We have established the charge theorem of the Hodge Complex Standard Model and resolved the status of its closed form. The electric charge $q_3 = F(n_1, \sigma)$ is a bijection from the 14 realized (n_1, σ) pairs to the 5 Standard Model charge eigenvalues. The three σ -lines carry charges 0, +5, −8, summing to the vacuum invariant $-1e$. Three structural properties of the charge on the 8×8 torus are established: even-sum, the $n_1 + n_2 = 4$ line containing all 5 charges, and the $n_1 + n_2 \equiv 2 \pmod{4}$ line carrying the entire vacuum charge.

The closed form of F does not exist in any of nine candidate algebraic families. The charge is the residue of a sorting operation on the 14-mode multiplet whose sort key is the zero state. The operation is stated explicitly, its four structural consequences are proved, and the reduction of Item 1.4 to Item 1.2 is established.

The charge theorem is therefore a corollary of the mass-label theorem. The framework's remaining task in the charge sector is to derive the $N/4$ mass labels from the zero state — Item 1.2 of the derivation list.

The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none. The charge table is one of those anchors; it is now understood as the output of a specific machine, not as a free input.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

References

1. S. Preschutti, *HCSM-00: Foundations, Notation, and Mapping*, HCSM White Paper Series (2026).
2. S. Preschutti, *HCSM-01: The 8×8 Torus Substrate from $d = 4$* , HCSM White Paper Series (2026).
3. S. Preschutti, *HCSM-02: The Hodge Complex on Λ* , HCSM White Paper Series (2026).
4. S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).
5. S. Preschutti, *HCSM-05: Orbit Decomposition and Inverse Participation*, HCSM White Paper Series (2026).
6. S. Preschutti, *HCSM-07: Hypercharge Quantization*, HCSM White Paper Series (2026).
7. S. Preschutti, *HCSM-11: The $5 + 9$ Split*, HCSM White Paper Series (2026).
8. S. Preschutti, *HCSM-13: The Processor T* , HCSM White Paper Series (2026).
9. S. Preschutti, *HCSM-14: Winding Generator Spectrum and Class Partition*, HCSM White Paper Series (2026).
10. S. Preschutti, *HCSM-20: The Motion Theorem*, HCSM White Paper Series (2026).
11. S. Preschutti, *HCSM-22: The Zero State*, HCSM White Paper Series (2026).
12. S. Preschutti, *HCSM-23: The Identification of the 14 Modes*, HCSM White Paper Series (2026).
13. S. Preschutti, *HCSM-25: The 4×4 Periodic Charge Density*, HCSM White Paper Series (2026).
14. S. Preschutti, *HCSM-26: The Observer Pair $\{W, t\}$* , HCSM White Paper Series (2026).
15. S. Preschutti, *HCSM-27: Klein Orbit Structure*, HCSM White Paper Series (2026).
16. S. Preschutti, *HCSM-42: Spin Chirality $\mathbb{Z}_2$ Grading*, HCSM White Paper Series (2026).
17. S. Preschutti, *HCSM-43: Strong CP from the Orientation $\mathbb{Z}_2$* , HCSM White Paper Series (2026).
18. Particle Data Group (S. Navas et al.), Review of Particle Physics, *Phys. Rev. D* **110**, 030001 (2024).